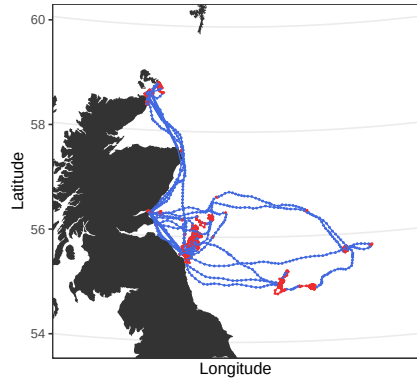


HMMs WITH NON-LINEAR AND RANDOM EFFECTS

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ANIMOVE 2025



HMM DATA REGULARISATION

Assumption: regular time intervals

As time between locations increases:

- ▶ animals can move farther = longer step lengths
- ▶ animals are more likely to change direction = less concentrated turning angles
- ▶ animals are more likely to switch between behaviours = higher between-state transition probabilities (i.e., lower state persistence)

Irregular data

Different forms of irregular data:

- ▶ Regular(-ish) time interval with some missing fixes
- ▶ Duty-cycled data (e.g., 1-minute for 4 hours, then 30-minute for 4 hours)
- ▶ Completely irregular locations (e.g., collected when an animal surfaces the water)

Irregular data

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Potential solutions:

- ▶ Split track at large gaps
- ▶ Fill missing locations with NAs
- ▶ Thin data to coarser scale resolution
- ▶ Use state-space model to impute locations

We would often use a combination of methods

Splitting track at large gaps:

- ▶ Segment an individual's track where there are large temporal gaps
- ▶ May want to do this in several situations (e.g., prior to filling missing locations or using a state-space model)
- ▶ Function available in moveHMM to do this (maybe others?)

Filling missing locations with NAs:

- ▶ Retain row for missing locations (needed for HMM), but fill observations with NAs
- ▶ Good solution for small gaps
- ▶ Can potentially be used for large gaps, but this will increase computational costs
- ▶ *If there are spatial covariates on the transition probabilities: split at large gaps first*
- ▶ Function in `adehabitatLT` to do this

Irregular data: solutions

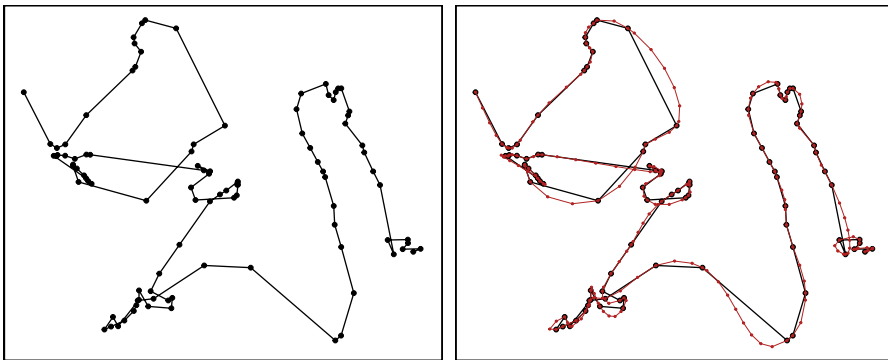
Thin data to coarser resolution:

- ▶ Resample track to a coarser resolution that covers the finer-scale data
- ▶ Useful for duty-cycled data
- ▶ Probably difficult if the data is very irregular

Irregular data: solutions

Impute locations with a state-space model:

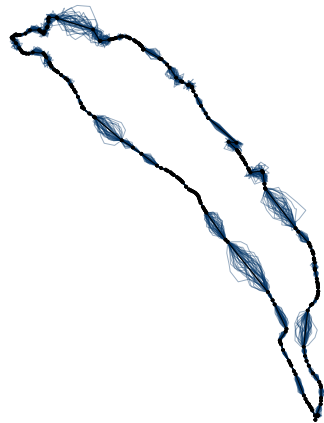
- ▶ Fit (continuous-time) state-space model to data and predict missing locations
- ▶ Useful for very irregular data or data with measurement error
- ▶ Does not perform well over long gaps
- ▶ Introduces new form of error from the model predictions



Irregular data: solutions

Impute locations with a state-space model:

- ▶ Possible to do this in several packages (aniMotum, ctm, momentuHMM)
- ▶ Can partially account for predicted location error with multiple imputation in momentuHMM



NON-LINEAR AND RANDOM EFFECTS

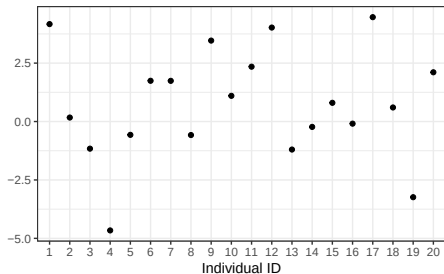
Specify parameters with flexible mixed effect model:

$$\eta = \mathbf{X}\beta + \mathbf{U}\alpha, \quad \text{where } \alpha \sim MVN\left(\mathbf{0}, \frac{1}{\lambda}\mathbf{S}^{-}\right)$$

- ▶ $\mathbf{X}$ is a model matrix for the fixed effects β
- ▶ $\mathbf{U}$ is a model matrix for the random effects α
- ▶ λ is a smoothing parameter and $\mathbf{S}$ is a penalty matrix
- ▶ Together, λ and $\mathbf{S}$ define the random effect covariance matrix

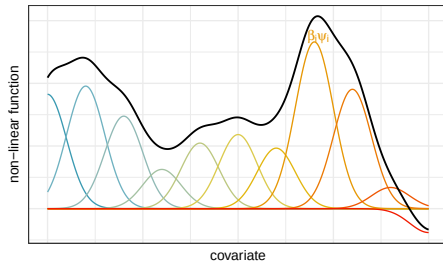
Choice of penalty and λ interpretation

Random slopes/intercepts:



- ▶ **S** is identity matrix
- ▶ λ = inverse of variance
- ▶ bigger λ = less variability

Penalised splines:



- ▶ **S** is specific to basis functions
- ▶ λ = wiggleness of the function
- ▶ bigger λ = straighter line

Special case 1: capturing inter-individual variability

- ▶ **Option 1:** fit separate models
 - difficult to combine/compare
 - assumes all individuals are completely independent
- ▶ **Option 2:** include “ID” as covariate
 - many parameters to be estimated
 - assumes all individuals are completely independent
- ▶ **Option 3:** include “ID” as a *random effect*
 - individuals deviate from some population mean
 - good compromise between complexity and realism
 - note: can also be used for any other relevant variable (e.g., year)

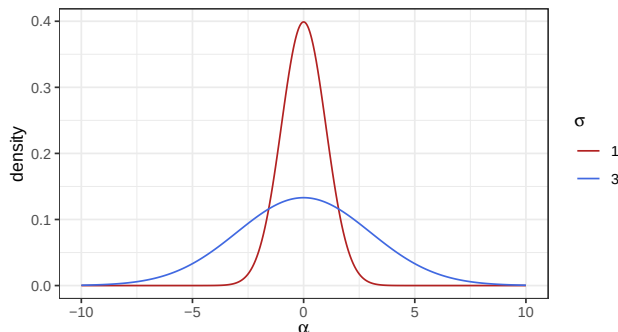
Special case 1: capturing inter-individual variability

Assumes individual-level parameters come from a Gaussian distribution:

$$\alpha_i \sim N(0, \sigma^2)$$

where

- ▶ α_i = deviation from population mean for individual i
- ▶ σ^2 = variance/spread around population mean



Special case 1: capturing inter-individual variability

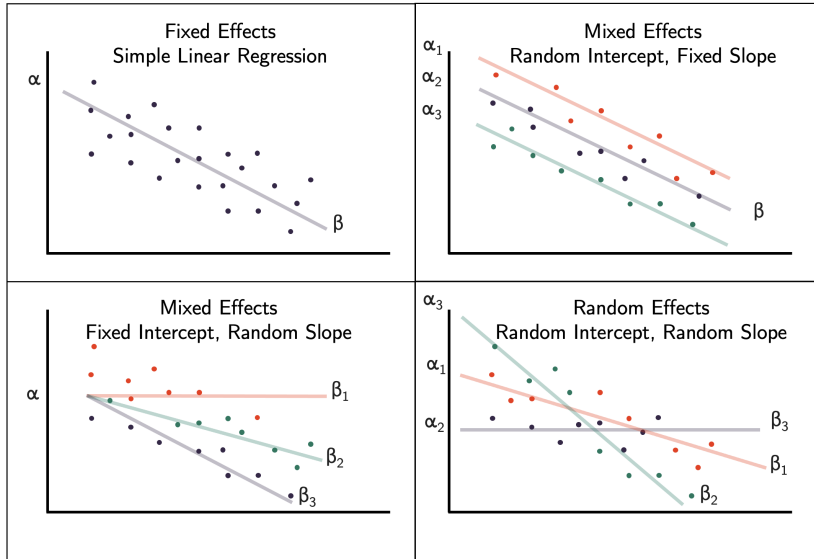
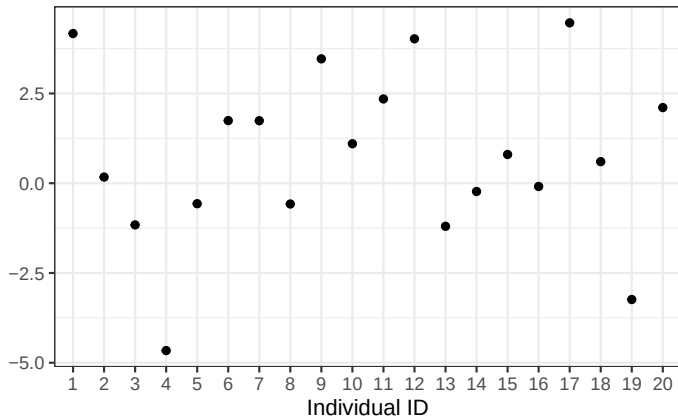


Figure credit: Steve Midway (note, here α is the intercept!)

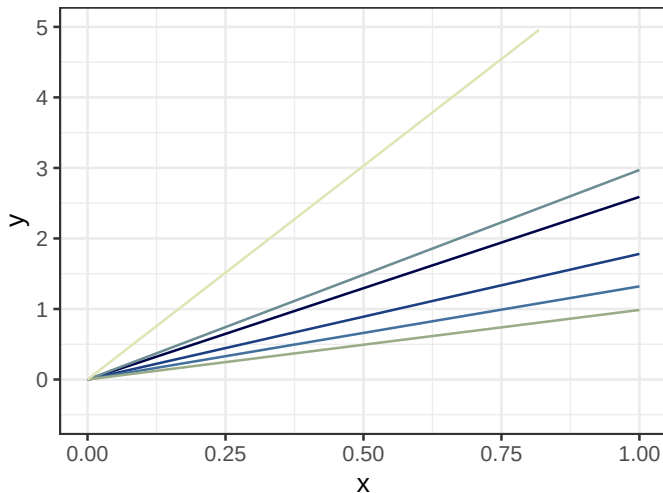
Special case 1: capturing inter-individual variability

Random intercept: $\eta_i = (\beta_0 + \alpha_i) + \beta_1 x_1 + \dots \beta_p x_p$



Special case 1: capturing inter-individual variability

Random slope: $\eta_i = \beta_0 + (\beta_1 x_1 + \alpha_i) + \dots \beta_p x_p$



Special case 2: non-linear covariate effects

Penalised splines are formed from a linear combination of basis functions

$$\alpha \sim MVN \left(\mathbf{0}, \frac{1}{\lambda} \mathbf{S} \right)$$

where

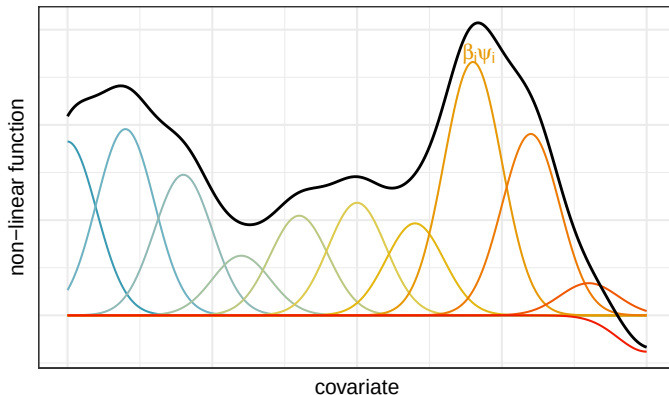
- ▶ α_i is the basis weight
- ▶ $\mathbf{S}$ is a penalty matrix specific to the basis functions chosen
- ▶ λ controls smoothness (i.e., variance of α)

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \\ \vdots \end{pmatrix} = \begin{pmatrix} 1 & x_{11} \\ 1 & x_{12} \\ 1 & x_{13} \\ \vdots & \vdots \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} \psi_1(x_{21}) & \psi_2(x_{21}) & \psi_3(x_{21}) & \cdots \\ \psi_1(x_{22}) & \psi_2(x_{22}) & \psi_3(x_{22}) & \cdots \\ \psi_1(x_{23}) & \psi_2(x_{23}) & \psi_3(x_{23}) & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \vdots \end{pmatrix}$$

Special case 2: non-linear relationships

Creates a non-linear function

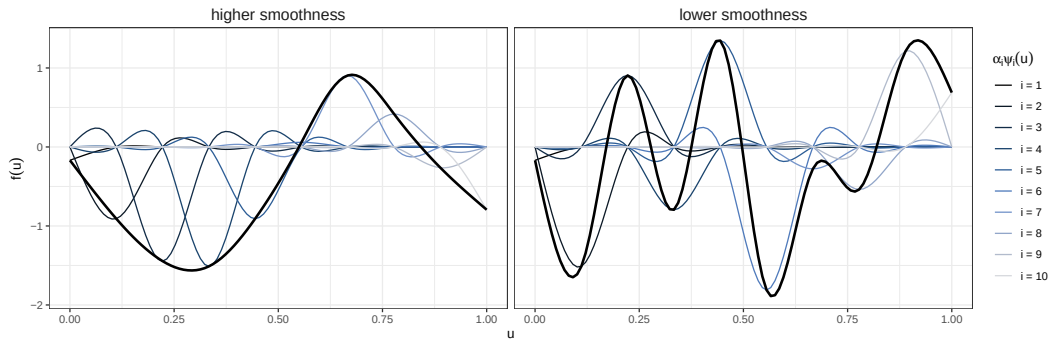
- ▶ can choose the number and type of basis functions
- ▶ cyclic, multivariate, etc.



Special case 2: non-linear relationships

λ controls how similar basis functions are (i.e., smoothness)

- ▶ not controlled by the number of functions (unless it is too few)



Model specification with mgcv¹

mgcv can create the model matrices **X** and **U**, with great flexibility:

- ▶ use `s( )` to specify a smooth term
- ▶ `bs` argument chooses basis function (i.e., type of smoother)
- ▶ `k` argument chooses the number of basis functions (upper limit to wiggleness, not needed for random slopes or intercepts)

Example

```
~ x + s(y, bs = "cr", k = 10) + s(ID, bs = "re")
```

- ▶ linear effect of `x`
- ▶ cubic spline for `y`
- ▶ random effect for `ID`

¹Wood (2017), Generalized Additive Models: an Introduction with R

Model specification with mgcv: important cases

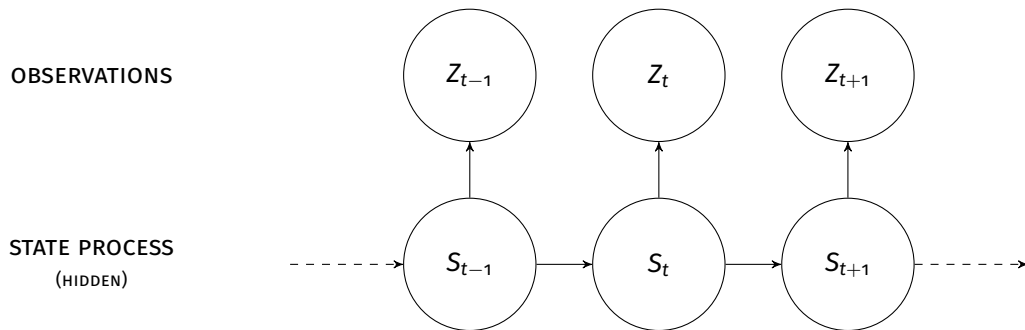
Model term	mgcv syntax
Random intercept	<code>s(animalID, bs = "re")</code>
Random slope	<code>s(covariate, animalID, bs = "re")</code>
Non-linear covariate effect	<code>s(covariate)</code>
Cyclic spline	<code>s(covariate, bs = "cc")</code>
Non-linear (isotropic) interaction	<code>s(covariate1, covariate2)</code>
Non-linear (anisotropic) interaction	<code>te(covariate1, covariate2)</code>
Non-linear interaction (no main effect)	<code>s(covariate2, by = covariate1)</code>

HMMS

Reminder: HMM structure

Doubly-stochastic process:

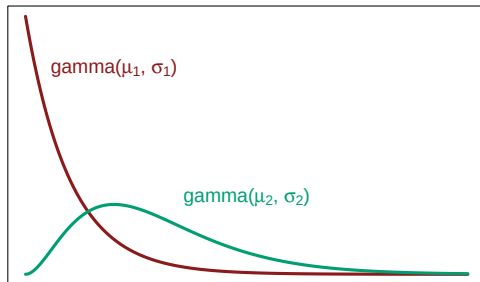
1. **observation process:** how the latent states give rise to the data;
2. **state process:** describes the state-switching dynamics.



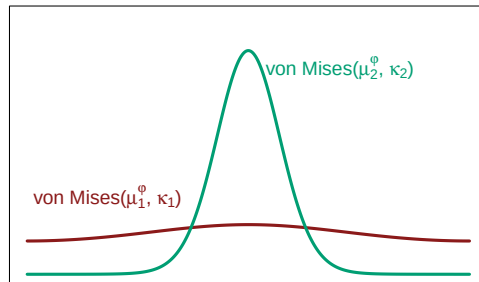
Reminder: HMM parameters

observation process: parameters of state-dependent distributions

step length



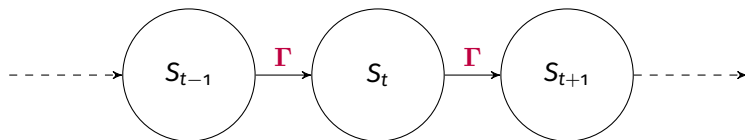
turning angle



Reminder: HMM parameters

state process: transition probabilities (describe probability of switching between states)

$$\mathbf{\Gamma} = \begin{pmatrix} \gamma_{11} & \gamma_{12} & \cdots & \gamma_{1K} \\ \gamma_{21} & \gamma_{22} & \cdots & \gamma_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{K1} & \gamma_{K2} & \cdots & \gamma_{KK} \end{pmatrix}, \quad \text{where } \gamma_{ij} = \Pr(S_t = j \mid S_{t-1} = i)$$



In Tutorial 2:

- ▶ Any parameter in the observation or state process can be modelled as a function of covariates

We can extend this to:

- ▶ Model these parameters with non-linear functions of covariates
- ▶ Include random effects (intercepts/slopes) on any parameter

Can be done in the `hmmTMB` package

The hmmTMB package

hmmTMB uses mgcv and TMB to implement flexible HMMs

- **Package benefits:** fast, many advanced features
- **For vignettes:** <https://github.com/TheoMichelot/hmmTMB>



Journal of Statistical Software

MMMMMM YYYY, Volume VV, Issue II.

<http://www.jstatsoft.org/>

hmmTMB: Hidden Markov Models with Flexible Covariate Effects in R

Théo Michelot
Dalhousie University

Abstract

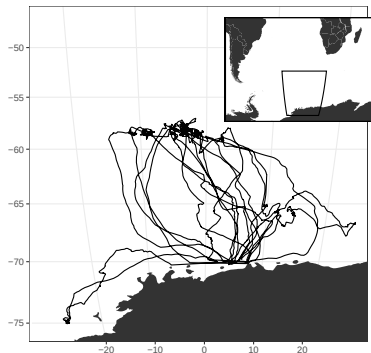
Hidden Markov models (HMMs) are widely applied in studies where a discrete-valued process of interest is observed indirectly. They have for example been used to model behaviour from human and animal tracking data, disease status from medical data, and financial market volatility from stock prices. The model has two main sets of parameters: transition probabilities, which drive the latent state process, and observation parameters, which characterise the state-dependent distributions of observed variables. One particu-

hmmTMB uses object-oriented programming

- ▶ Creates objects of different classes (i.e., data structures)
- ▶ Can run functions specific to the class
- ▶ For HMM: create observation object, Markov chain, and HMM

```
# Create object of class Class  
object <- Class$new()  
# Apply method() on object  
object$method()
```

Example: petrel movement data



Descamps et al. (2016), Movebank

Data:

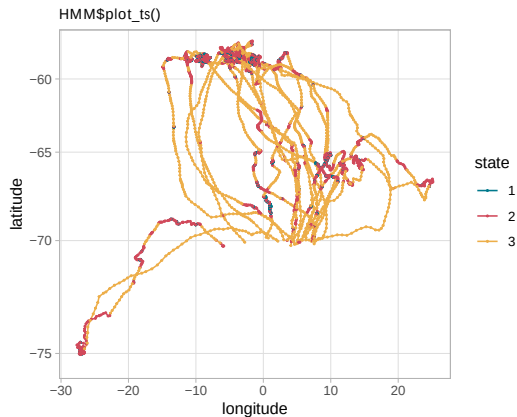
- ▶ 10 tracks (from 10 individual petrels)
- ▶ locations at 30-min resolution

Model:

- ▶ 3 states
- ▶ step length $\sim$ gamma
- ▶ turning angle $\sim$ wrapped Cauchy
- ▶ non-linear effect of distance to colony with individual-level random intercept

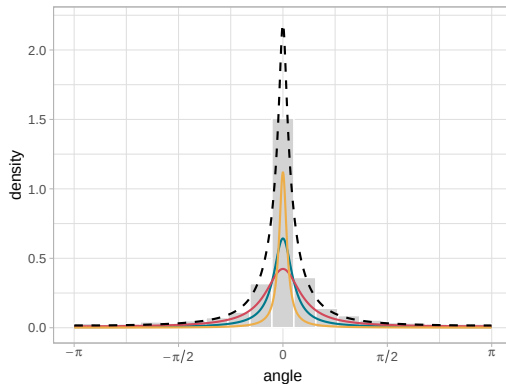
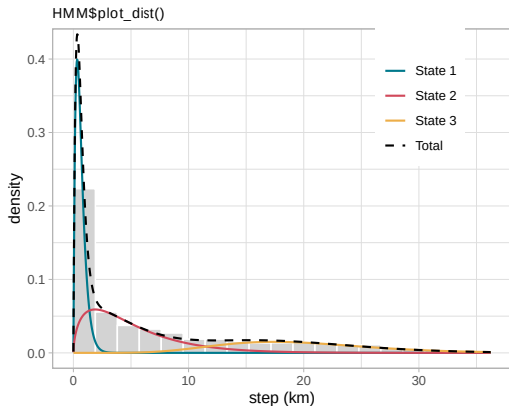
Petrel analysis results

The Viterbi algorithm gives us a classification of the observations into states



Petrel analysis results

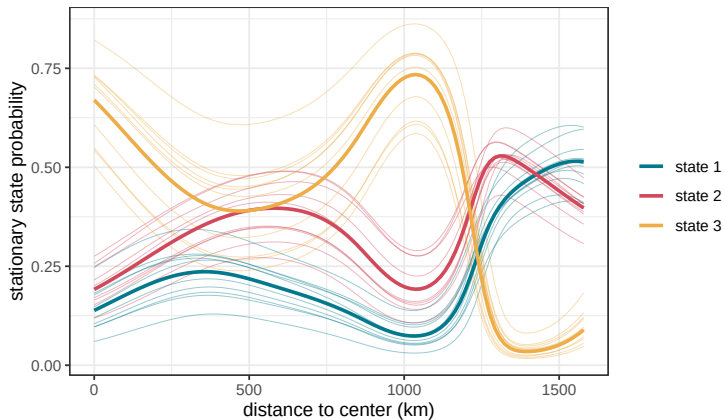
- ▶ State 1 = slow (“resting”, “foraging”?)
- ▶ State 2 = moderate speed + least directed (“searching for food”?)
- ▶ State 3 = fast + most directed (“transit”?)



Petrel analysis results

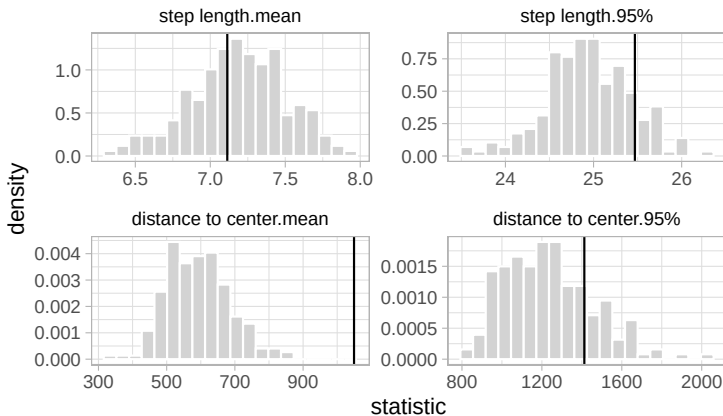
We can plot the estimated effects of the covariates on the state process.

- ▶ petrels tend to move fast up to 1000km from colony, then start foraging?



Simulation-based model checking

1. simulate from fitted model many times
2. compare some metrics for the simulated and observed data
3. discrepancies point to lack of fit



A word of caution

momentuHMM and hmmTMB make it possible to implement very complex models

- ▶ computational cost increases fast with number of states, number of covariates, etc
- ▶ numerical instability increases even faster
- ▶ estimating covariates on dynamics of unobserved process is challenging
- ▶ we can't include as many variables as in simpler regression models
- ▶ likely won't be able to perfectly meet every assumption

Suggested workflow:

1. start from covariate-free 2-state model
2. slowly increase complexity
 - add one state at a time
 - add one covariate at a time
3. stop if:
 - it becomes numerically unstable
 - interpretation becomes difficult

SSFs

SSFs with non-linear and random effects

An SSF with smooth/random effects takes the form,

$$\Pr(y_{it} = 1 \mid \mathbf{X}_t, \mathbf{Z}_t) = \frac{\exp(\beta^\top \mathbf{x}_{it} + \gamma^\top \mathbf{z}_{it})}{\sum_{n=0}^N \exp(\beta^\top \mathbf{x}_{nt} + \gamma^\top \mathbf{z}_{nt})}, \quad \gamma \sim \text{MVN} \left(\mathbf{o}, \frac{1}{\lambda} \mathbf{S} \right)$$

where

- ▶ $\mathbf{X}_t$ is the design matrix for the fixed effects β
- ▶ $\mathbf{Z}_t$ is the design matrix for the (normally distributed) **random effects** γ
- ▶ $\mathbf{S}$ is a penalty matrix and λ is a smoothing parameter

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where

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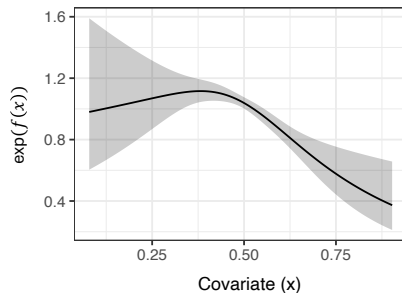
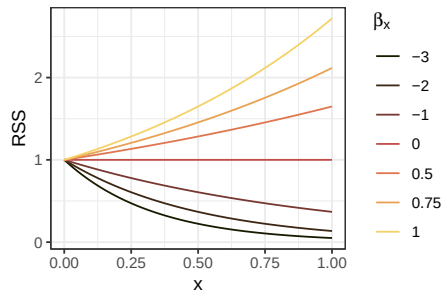
```
2 library(mgcv)
3
4 fit <- gam(cbind(times, stratum) ~
5           s(step) + s(angle) + s(cov1),
6           data = data,
7           family = cox.ph,
8           weights = obs,
9           method = "REML")
```

Non-linear covariate effects: **habitat**

Allow non-linear patterns of habitat selection

► **e.g.**, include smooth effect of x :

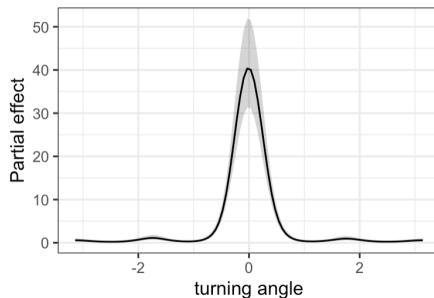
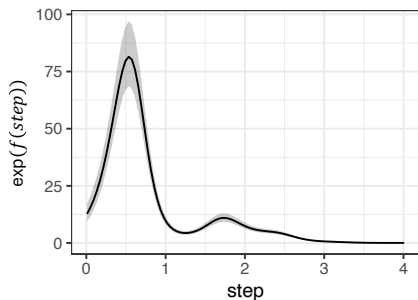
$$\beta_1 L_t + \beta_2 \log(L_t) + \beta_3 \cos(\theta_t) + \beta_4 x \rightarrow \beta_1 L_t + \beta_2 \log(L_t) + \beta_3 \cos(\theta_t) + f(x)$$



Non-linear covariate effects: **movement**

Can also be for more flexible movement kernels with non-parametric distributions:

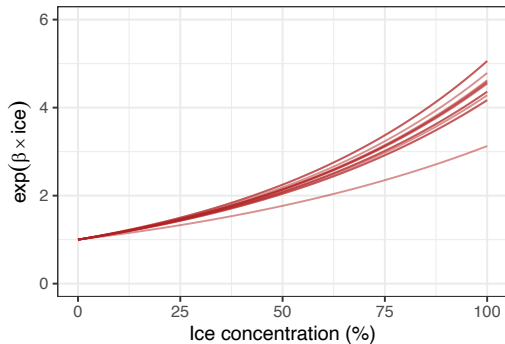
$$\beta_1 L_t + \beta_2 \log(L_t) + \beta_3 \cos(\theta_t) + \beta_4 \mathbf{x} \rightarrow f(L_t) + f(\theta) + \beta_4 \mathbf{x}$$



Random effects: slopes

SSF does not have an intercept so we can only specify **random slopes**

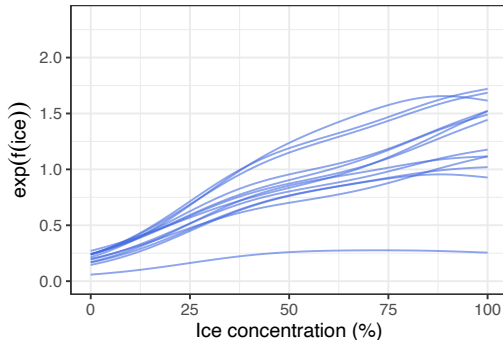
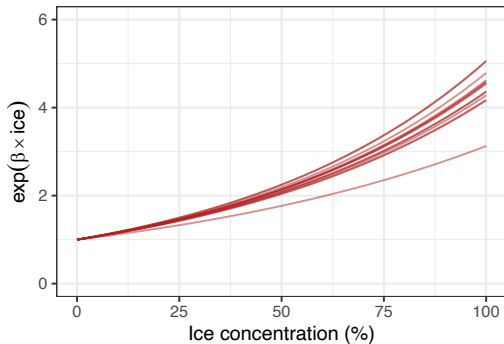
- e.g., relative selection or movement varies by individual



Random effects: slopes

SSF does not have an intercept so we can only specify **random slopes**

- e.g., relative selection or movement varies by individual



1. Prepare movement data (using `amt`)
 - Add random points and calculate movement variables
 - Interpolate spatial covariates
2. Fit model in `mgcv`
 - Implementation as a Cox PH model
3. Interpret model outputs using `gratia` and `mgcv`
 - Smooths visualisation
 - Derive relative selection strength and movement distributions
 - Note: difficult model checking for SSFs

- 🌐 Github repository
 - github.com/NJKlappstein/smoothSSF
- 📄 Klappstein, NJ., et al. 2024. Step selection functions with non-linear and random effects. *Methods in Ecology and Evolution*.